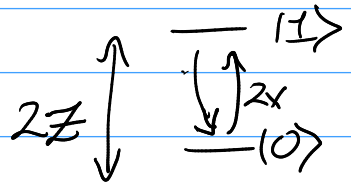
Spin Qubits (Quantum-Dots)

Review

$$\mathcal{H} = X\sigma_x + Y\sigma_y + Z\sigma_z = \begin{pmatrix} Z & X-iY \\ X+iY & -Z \end{pmatrix} \quad \text{in } |0\rangle/|1\rangle \text{ basis.}$$

$\langle 0|\mathcal{H}|0\rangle = Z$

$\langle 1|\mathcal{H}|1\rangle = -Z$

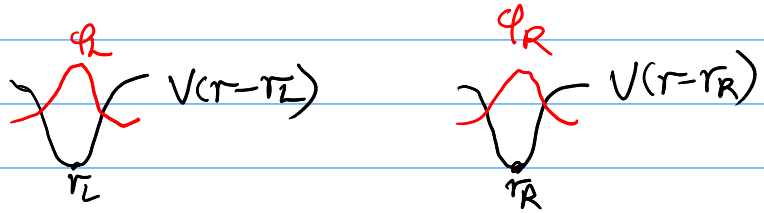


$\omega = \frac{2}{\hbar}\sqrt{X^2+Y^2+Z^2}$; if $Y = Z = 0$, $\omega_x = \frac{2}{\hbar}X$

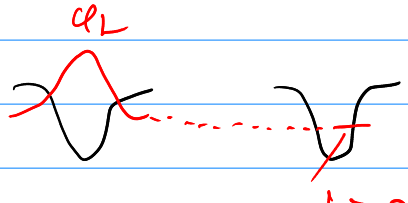
$2X$ gives energy/rate of state toggling $|0\rangle \leftrightarrow |1\rangle$

Charge Qubit (1 e^-, 2 dots)

A quantum dot is a '0D potential well' that traps e's or holes.

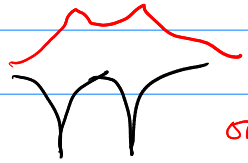


Tight-binding approximation: the basis eigenfunction that solves $H = \frac{-\hbar^2}{2m}\nabla^2 + V(r-r_L)$ solves $H = \frac{-\hbar^2}{2m}\nabla^2 + V(r-r_L) + V(r-r_R)$ with a small perturbation in energy.



i.e. we want the states $|\phi_L\rangle$ $|\phi_R\rangle$ to be weakly interacting or 'tightly bound'.

i.e. NOT



this is NOT a perturbation or linear comb. of $|\phi_L\rangle$ & $|\phi_R\rangle$

c.f. LCAO $\sum_{bs} = c$

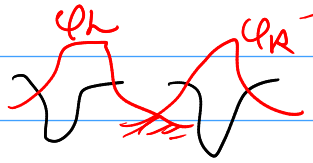
$\langle \varphi_L | H | \varphi_L \rangle = U_L$

$\langle \varphi_R | H | \varphi_R \rangle = U_R$

$H|\varphi_R\rangle = \left(\underbrace{-\frac{\hbar^2}{2m}\nabla^2 + V(r-r_R)} + V(r-r_L)\right)|\varphi_R\rangle$

$= (U_R + V(r-r_L))|\varphi_R\rangle$

$\langle \varphi_L | H | \varphi_R \rangle = (U_R + V(r-r_L)) \cdot \underbrace{\langle \varphi_L | \varphi_R \rangle} = -t_c$



$U < 0$ (attractive)

$-\frac{\hbar^2}{2m}$

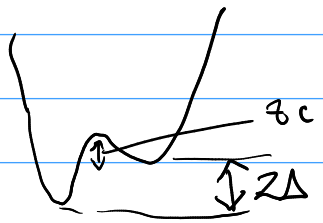
neg. $t_c > 0$

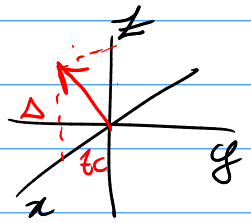
$2\Delta = U_R - U_L$

Qubit Hamiltonian. in $|\varphi_L\rangle / |\varphi_R\rangle$ basis.

$$H = \begin{pmatrix} \Delta & -t_c \\ -t_c & -\Delta \end{pmatrix}$$

$$= \Delta \sigma_z - t_c \sigma_x$$

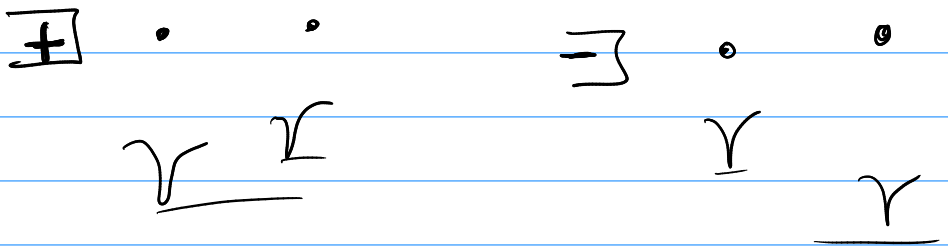




$\Delta \sim 0 - 1000$

$t_c \sim 1$

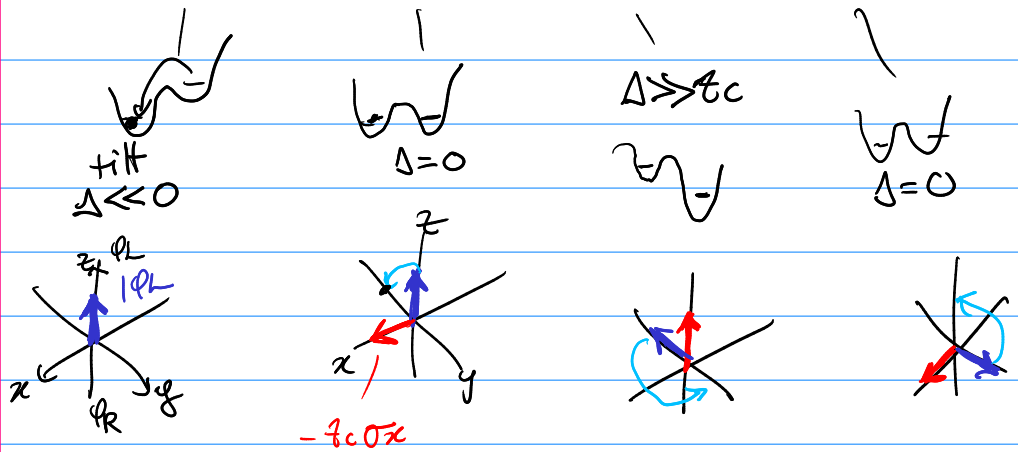
Typically control Δ via gate potentials.



Example operation.

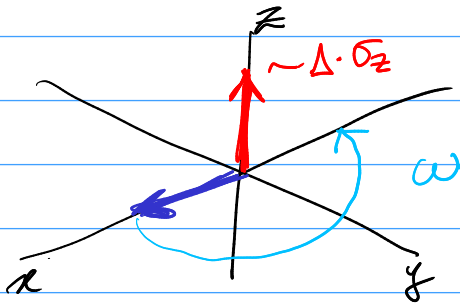
$|\varphi_L\rangle \rightarrow \boxed{X\pi/2} \rightarrow \boxed{Z\sigma} \rightarrow \boxed{X\pi/2} \rightarrow \boxed{\alpha}$

Key: state, rot. axis



Decoherence.

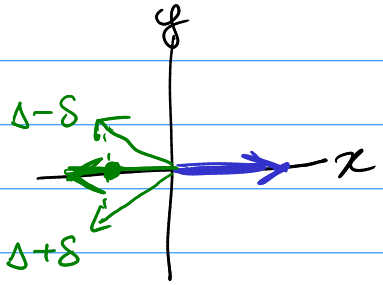
Consider a Z_π-rotation.



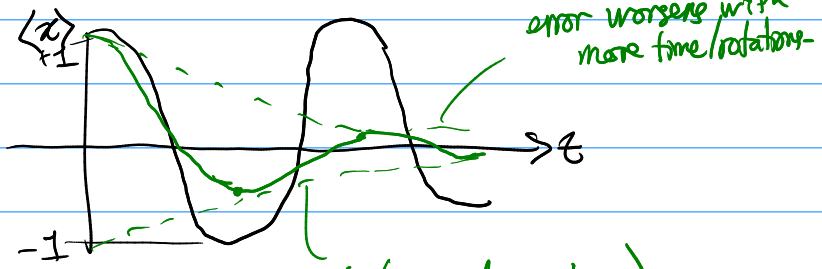
$\omega \sim \frac{2\cdot\Delta}{\hbar} \Rightarrow t_\pi \sim \frac{2\pi}{\omega}$

What if there's stochastic noise?

$\Delta \to \Delta + \delta$

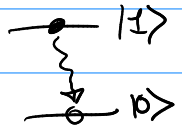


Can try to "fix" this decoherence by guessing the noise δ.
ie/ Hahn echo, CPMG etc.

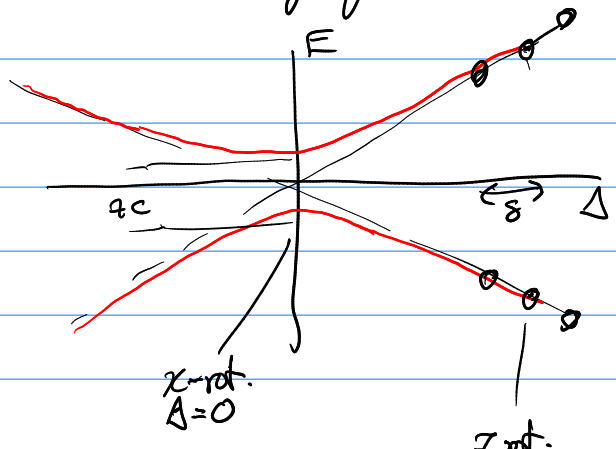


$T_2^* \leq T_2 \leq T_1$

$\boxed{2T_1 \geq T_2^*}$



We control charge qubits via Δ. So plot eigenvalues (energy) as func. Δ.



$E = \pm 2\sqrt{x^2+y^2+z^2}$

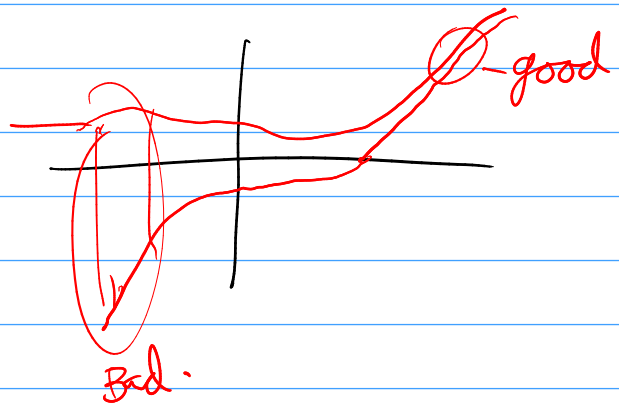
$= \pm 2\sqrt{\varepsilon_c^2+\Delta^2}$

ie/ variation in E due to noise δ

ie.

$\sim \frac{\partial(E_{upper} - E_{lower})}{2\Delta}$

$\Rightarrow$ slope in the energy difference must be small.



Charge Qubits.

$T_2^* \sim$ 10's ns.

Singlet - Triplets. (2e⁻s, 2 dots)

Add 2e⁻s, 2 dots.

1st e⁻

$(|\Phi\rangle \otimes |S\rangle) \otimes (|\Phi\rangle \otimes |S\rangle)$ $\}$ $(2\times2)^2 = 16$ states.

L or R ; ↑ or ↓

Can we cheat?

Pauli Exclusion principle! Fermions must form an anti-symmetric wavefunction.

Take $\hat{P}$ to be the exchange operator

i.e. $\hat{P}|\downarrow\downarrow\rangle = |\downarrow\downarrow\rangle$ - eigenvalue is $+1 \Rightarrow$ symmetric

Similarly. $\hat{P}|\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle$ " " " "

2 more states (i.e. 2 spins $\Rightarrow$ $2\times2=4$ spin states)

$\hat{P}|\uparrow\downarrow\rangle = |\downarrow\uparrow\rangle$ $\}$ i.e. NOT eigenstate.
$\hat{P}|\downarrow\uparrow\rangle = |\uparrow\downarrow\rangle$ $\}$

$\hat{P}\left(\frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}\right) = \frac{|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle}{\sqrt{2}} \Rightarrow$ eigenvalue is $+1$

$\hat{P}\left(\frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}\right) = \frac{|\downarrow\uparrow\rangle - |\uparrow\downarrow\rangle}{\sqrt{2}} = -\left(\frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}\right) \Rightarrow$ eigenvalue is -1

antisymmetric

Relabel.

$$T_- = |\downarrow\downarrow\rangle$$
$$T_+ = |\uparrow\uparrow\rangle$$
$$T_0 = \frac{1}{\sqrt{2}}\cdot(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$
$$S_0 = \frac{1}{\sqrt{2}}\cdot(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

} Symmetric Triplets.

} Antisymmetric Singlet.

So we can eliminate states...

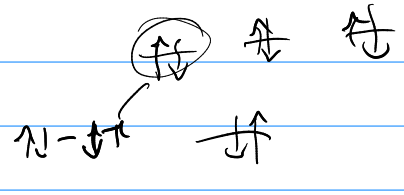
$$\hat{P}(|\varphi_L\varphi_L\rangle \otimes T_-) = |\varphi_L\varphi_L\rangle \otimes |\downarrow\downarrow\rangle \Rightarrow +1 \text{ eigenvalue} \Rightarrow \text{symmetric.}$$
($\hat{P}$: $\varphi_L \leftrightarrow \varphi_L$ gives $+1$; T_- gives $+1$)

$$\hat{P}\left(\frac{1}{\sqrt{2}}(|\varphi_L\varphi_R\rangle + |\varphi_R\varphi_L\rangle) \otimes |T_-\rangle\right) \Rightarrow +1 \text{ eigenvalue}$$
($+1$, $+1$)

$$\hat{P}\left(\frac{1}{\sqrt{2}}(|\varphi_L\varphi_R\rangle - |\varphi_R\varphi_L\rangle) \otimes |T_-\rangle\right) \Rightarrow -1 \text{ eig. value}$$
(-1, $+1$)

In fact all triplets must live on separate dots. via the $\frac{1}{\sqrt{2}}(|\varphi_L\varphi_R\rangle - |\varphi_R\varphi_L\rangle)$ state (unless another DOF like orbital is introduced). So. $T_-(1,1)$, $T_+(1,1)$, $T_0(1,1)$ are valid (usually we drop e⁻ number or spatial state). For singlets.

$$\hat{P}(|\varphi_L\varphi_L\rangle \otimes |S_0\rangle) \Rightarrow -1 \text{ eig. value.}$$
($+1$, -1)



$$\hat{P}\left(\frac{(|\varphi_L\varphi_R\rangle + |\varphi_R\varphi_L\rangle)}{\sqrt{2}} \otimes |S_0\rangle\right) \Rightarrow -1 \text{ eig. value.}$$
($+1$, -1)

Abbreviate these states. S_{20}, S_{11}, S_{02}. (both left dot) Usually energy to shift from $S_{20} \leftrightarrow S_{02}$ is large. So we usually stick with

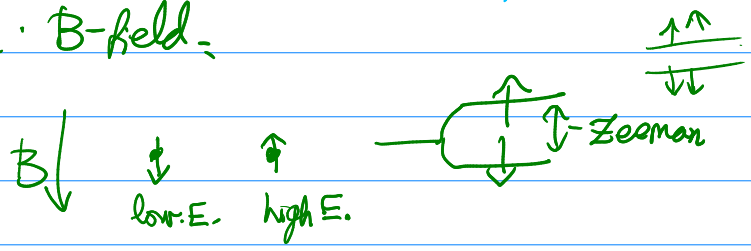
$T_-, T_+, T_0, S_{11}, S_{02}$

$$H_{BT} = \begin{array}{c} \begin{matrix} S_{02} & S_{11} & T_0 & T_- & T_+ \end{matrix} \\ \begin{pmatrix} -\Delta & -t_c & 0 & 0 & 0 \\ -t_c & \Delta & \Delta B_z & 0 & 0 \\ 0 & \Delta B_z & \Delta & 0 & 0 \\ 0 & 0 & 0 & \Delta - B_z & 0 \\ 0 & 0 & 0 & 0 & \Delta + B_z \end{pmatrix} \end{array} \begin{matrix} S_{02} \\ S_{11} \\ T_0 \\ T_- \\ T_+ \end{matrix}$$

2Δ is E-splitting between $(1,1)$ & $(0,2)$ i.e. $\downarrow\uparrow$ vs. $\uparrow\downarrow$

t_c tunnel couples $(1,1) \leftrightarrow (0,2)$, i.e. $S_{11} \leftrightarrow S_{02}$, NOT $T_{+,-,0}$ - forbidden. P.E.P.

Apply B-field:



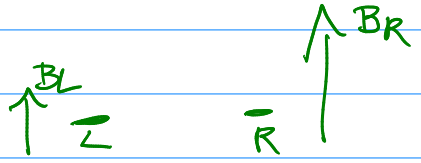
$$E_z = \frac{-g \mu_e \cdot B}{2}$$

omit this for clarity - use B_x, B_y, B_z

$T_0, S_{11}, S_{02}, |\downarrow\uparrow\rangle$ & $|\uparrow\downarrow\rangle$

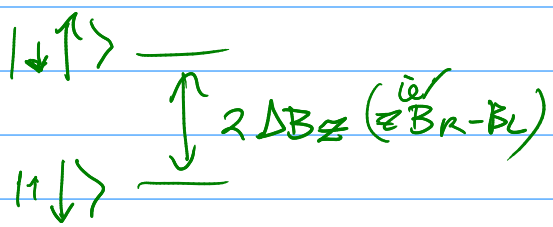
$m_z = 0$

But if B-field is locally different -



so. $|\uparrow\downarrow\rangle$ & $|\downarrow\uparrow\rangle$ are split -

i.e.



But notice -

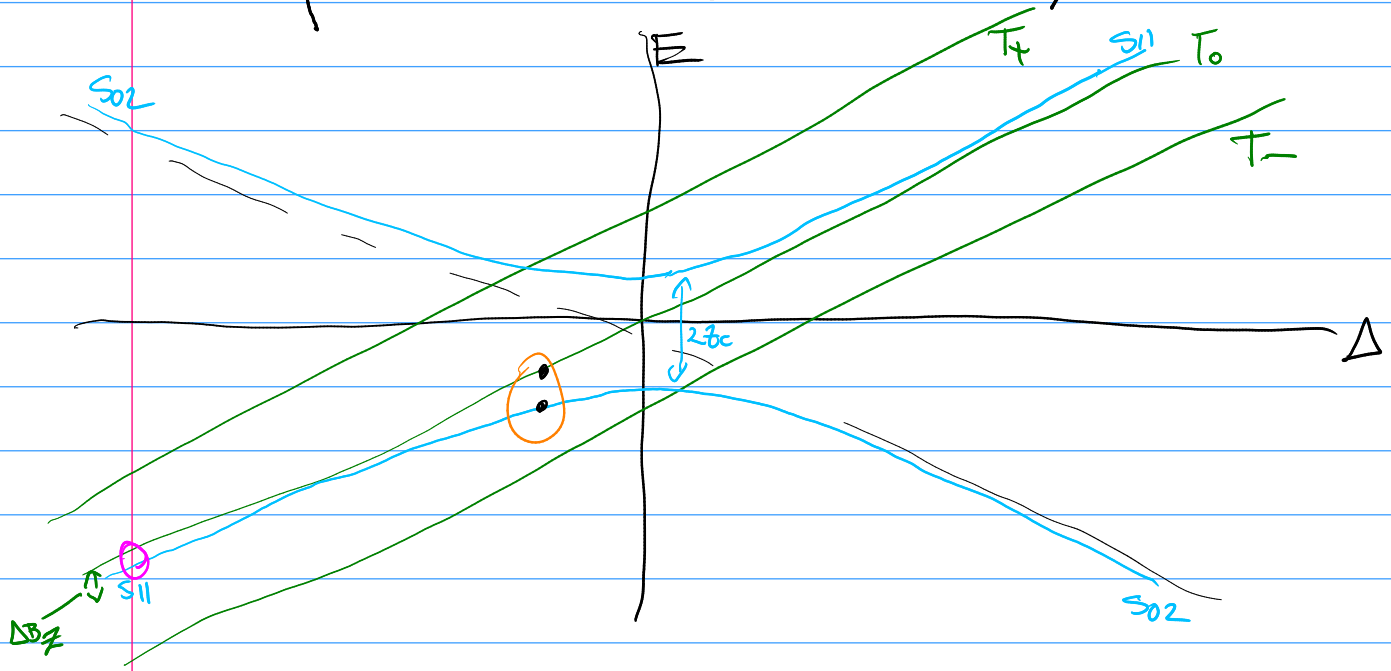
$$|\uparrow\downarrow\rangle = \frac{|S_0\rangle + |T_0\rangle}{\sqrt{2}}$$

$$|\downarrow\uparrow\rangle = \frac{|T_0\rangle - |S_0\rangle}{\sqrt{2}}$$

} i.e. the splitting between $\uparrow\rangle$ & $|\downarrow\rangle$ on Bloch sphere.

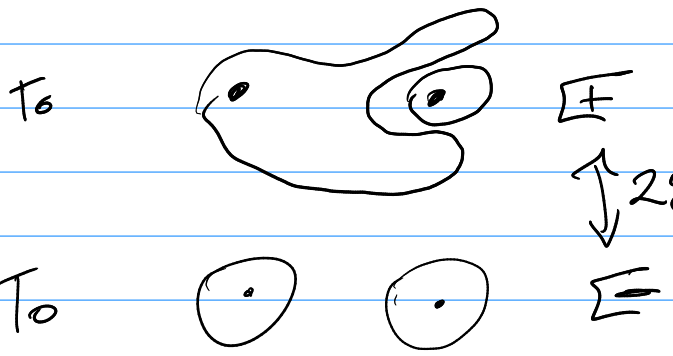
$\Rightarrow \sigma_x$ - like -

$$H_{BT} = \begin{array}{c} \begin{matrix} S_{02} & S_{11} & T_0 & T_- & T_+ \end{matrix} \\ \begin{pmatrix} -\Delta & -t_c & 0 & 0 & 0 \\ -t_c & \Delta & \Delta B_z & 0 & 0 \\ 0 & \Delta B_z & \Delta & 0 & 0 \\ 0 & 0 & 0 & \Delta - B_z & 0 \\ 0 & 0 & 0 & 0 & \Delta + B_z \end{pmatrix} \end{array} \begin{matrix} S_{02} \\ S_{11} \\ T_0 \\ T_- \\ T_+ \end{matrix}$$



Focus on S_{11} & T_0

$2J = \Delta + \sqrt{t_c^2 + \Delta^2}$ – J is singlet-triplet splitting. a.k.a. exchange interaction



Eff. Hamiltonian – S_{11}/T_0 basis.

$$H = \begin{pmatrix} J & \Delta B_z \\ \Delta B_z & -J \end{pmatrix} = J\sigma_z + \Delta B_z \cdot \sigma_x$$

x-rot. $\delta = 0$, $\Delta \to -\infty$, $\delta \approx -\Delta + \sqrt{\Delta^2} = 0$

z rot. $\delta \gg \Delta B_z$.

$\delta \ll \Delta B_z$  x-rot

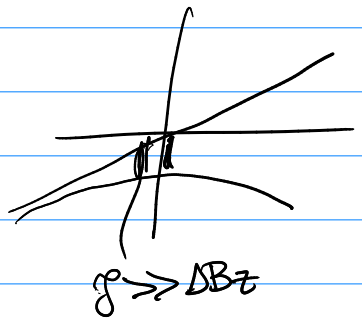
$\delta \gg \Delta B_z$  z-rot

Pros

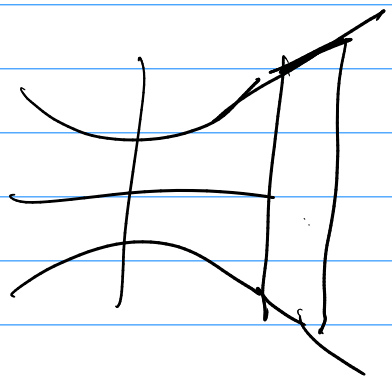
- In S/ϕ_0 basis, immune global B-field noise.
- Immune to charge noise - for x-rot.

Con

- q-noise dominates for z-rot.

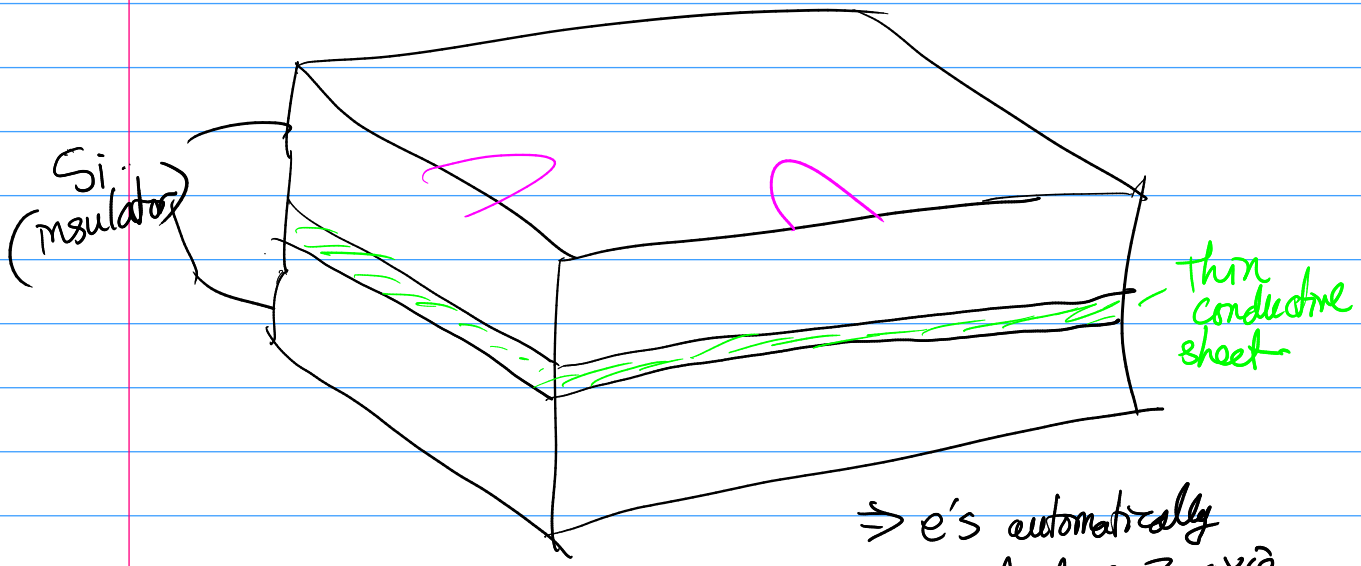


vs.



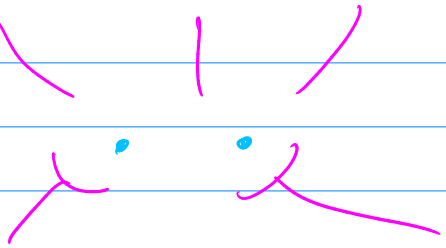
Platforms

Earnshaw Thm. - Static E-fields cannot confine. (@ / no minima).

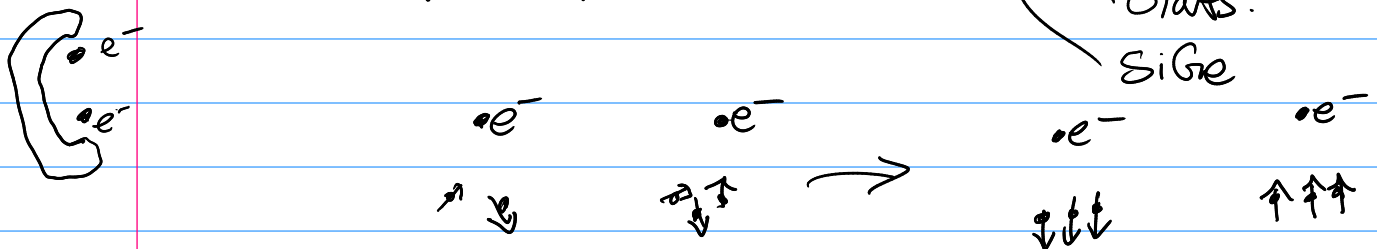


⇒ e's automatically confined on z-axis.

2DEG

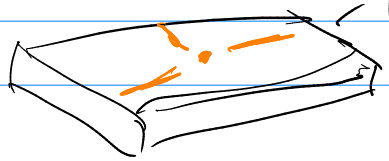


Gate-defined Quantum Dots - CMOS, GaAs, SiGe



Dynamic nuclear spin polarisation - (2008, 2012)

Donor-defined Q-dot.



P in Si, Pauli ⇒ confinement for free.

e⁻ lives in valence band.

- STM-defined -
- Ion-implantation -

